

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester II (NEP) Examination, 2024-25

MATHEMATICS**Paper Code: MATMJ21 / MATMN21 : Algebra**

Time: 2.5 Hours

Maximum Marks: 70

REF NO. CECNF/N/2024-25/23

*Instructions: Answer **five** questions in all including Question No. 1 which is compulsory. All questions carry equal marks (14 marks each).*

Question 1. (Compulsory, 14 Marks Total)

- (a) Express $-1 - \sqrt{3}i$ in modulus-amplitude form. [2]
- (b) If $S = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\}$, then determine infimum of S . [2]
- (c) Find all the roots of equation $Z^3 = -8i$, where $Z \in \mathbb{C}$ and $i = \sqrt{-1}$. [2]
- (d) State the fundamental theorem of Algebra. Determine all the roots of polynomial $p(x) = x^3 - 5x^2 - 16x + 80$. [1+1=2]
- (e) Define a relation R on $\mathbb{Z} - \{1\}$ by declaring that xRy if x and y have a common factor other than 1. Check whether R is an equivalence relation. [2]
- (f) Let G be a cyclic group and a (eqe) be an element of G . If a generates G , then show that order of G is same as order of a . [2]
- (g) Let $n \in \mathbb{Z}$ be a positive integer. Recall the congruence relation ' $\equiv$ ' defined on $\mathbb{Z}$. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$ for some a, b, c, d in $\mathbb{Z}$, then verify whether $ac \equiv bd \pmod{n}$. [2]

Question 2.

- (a) Use Euclidean algorithm to show that: if $a, b \in \mathbb{Z} - \{0\}$, then there exist $x, y \in \mathbb{Z}$ such that $ax + by = \gcd\{a, b\}$. [7]
- (b) Solve the following equation in the set of integers: $237x + 81y = 15$. [7]

Question 3.

- (a) Let S be the set of all real numbers except -1 . Define an operation ' $*$ ' on S by $a * b = a + b + ab$. Examine whether S is a group with the operation ' $*$ '. Is it an Abelian group? Justify. [8]
- (b) For $n = 12$ consider $\mathbb{Z}_n$. For each divisor m of n , write down an element $a \in \mathbb{Z}_n$ such that order of a is m . [6]

Question 4.

- (a) Define subgroup of a group. Prove that a non-empty subset H of a group G is a subgroup if and only if $a, b \in H \implies ab^{-1} \in H$. [7]
- (b) Show that a subgroup of a cyclic group is always cyclic. [7]

Question 5.

- (a) Define centre of a group and centralizer of an element. Consider the group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ which satisfies the following multiplication rules:

$$i^2 = j^2 = k^2 = -1, \quad ij = k, \quad jk = i, \quad ki = j, \quad ji = -k, \quad kj = -i, \quad ik = -j.$$

Find the centre of Q_8 and centralizer of i . [6]

- (b) Does the group of symmetries of a square contain an element of order six? Justify your answer. [8]

Question 6.

- (a) Determine the roots of the polynomial $x^3 + 10x + 5$ in $\mathbb{Z}$. What can you say about its roots in $\mathbb{Q}$? [7]
- (b) Find all the roots of polynomial $x^6 - 1 = f(x)$. Can you use Newton's method to find some roots of this polynomial? Justify your answer. [7]

Question 7.

- (a) Define a binary operation on the set of even integers which is different from addition, subtraction or multiplication. [2]
- (b) Find a solution of the equation $ax = b$ in S_3 , where

$$a = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}.$$

[3]

- (c) Give an example of a group which is not cyclic, but its every proper subgroup is cyclic. [2]
- (d) Let G be a group and $a, b \in G$. Prove the following:
- (i) $(ab)^{-1} = b^{-1}a^{-1}$
- (ii) $(ab)^{-1} = a^{-1}b^{-1} \iff G$ is Abelian.

[2]

(e) Define order of an element of a group. Let $O(a) = n$, show the following:

- (i) $a^m = e \iff m = nq$, where $q \in \mathbb{Z}$
- (ii) $O(a^s) \leq n$, where $s \in \mathbb{Z}$
- (iii) If p is prime to n , then $O(a^p) = n$.

[1+1+2=4]